

Arush Gupta

+91 7754843033 | arushgupta1905@gmail.com | linkedin.com/in/arushgupta | github.com/redpopcorn

EDUCATION

KIET Group of Institutions

Bachelor of Technology in Computer Science & Engineering

Ghaziabad, India

Aug 2022 – May 2026

CERTIFICATIONS

AWS Certified AI Practitioner

AWS Certified Cloud Practitioner

PROJECTS

AgentVault – Multi-Tenant AI Agent Orchestration Platform | *LangGraph, Redis, PostgreSQL, TypeScript* github.com/redpopcorn/AgentVault

- Designing and building a multi-tenant platform for orchestrating AI agent workflows, with tenant-isolated execution and state management via LangGraph
- Implemented core auth and access-control layer: JWT-based authentication, role-based access control (RBAC), and multi-tenant database schema in PostgreSQL
- Architected Redis-backed state and session management to support concurrent multi-agent orchestration across tenants
- Structured the system with an “understand-then-generate” development discipline to keep the schema and auth layer production-ready before extending to agent orchestration logic

OpenClaw – AI Coding Assistant (CLI + Telegram) | *TypeScript, Bun, OpenRouter API, Telegram Bot API, IMAP/SMTP*

- Built a multi-mode AI coding agent (Agent, Plan, Ask) that reads, edits, and executes commands directly against a codebase, with staged, interactive, and step-by-step approval workflows for safe autonomous changes
- Integrated email (IMAP/SMTP) and Telegram bot interfaces so the agent could be operated and approved remotely, with owner-only authorization
- Implemented codebase-aware Q&A combining local code search with external web research via the Firecrawl API
- Designed a modular tool architecture (file read/write/delete, shell execution, email, web research) routed through a single agent loop with action tracking

LearnHub – Full-Stack Course Selling Platform | *Node.js, Express, MongoDB, React, Next.js, bcrypt, JWT, Razorpay* github.com/redpopcorn/LearnHub

- Built a full-stack course marketplace with separate Student and Admin roles – admin can create, edit, and publish courses; students browse, purchase, and manage enrollments
- Implemented secure authentication with bcrypt password hashing, JWT session tokens, and refresh token rotation for persistent login
- Integrated Razorpay payment gateway with webhook signature verification for reliable order confirmation and purchase state management in MongoDB
- Built RESTful API with Express covering auth, course CRUD, enrollment, and admin dashboard; used Next.js SSR for SEO-optimized public-facing pages

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, SQL

Backend: Node.js, Express.js, REST API Design, JWT Auth, bcrypt, WebSockets

GenAI: LangChain, LangGraph, OpenAI API, Prompt Engineering, RAG Pipelines, Embeddings

Frontend: React, Next.js, HTML5, CSS3

Databases: PostgreSQL, MongoDB, Redis

Tools & Infra: Git, GitHub Actions, Docker, Vercel, Postman, Railway

RELEVANT COURSEWORK & EXTRA

Coursework: Data Structures & Algorithms, Database Management Systems, Operating Systems, Computer Networks, Object Oriented Programming

Open Source & Projects: Active GitHub profile with consistent project contributions – github.com/redpopcorn

Interests: Building AI-powered products, backend system design, exploring large language model applications and agentic workflows